

Course: BCA.

Semester/Year: Vth/IIIrd

Session: 2025-26

Subject Name & Code: Soft Computing [BCAC0028]

Section: B & J

Maximum Marks: 10

Assignment 1

Submission Date: 22/09/2025

Important Instructions:

- All answers must be **handwritten** on **A4-sized paper** using only **blue or black ink pens** for writing.
- Clearly write the **question number** and the **question** before providing the answer.
- Include **neat, accurate, and clearly labeled diagrams** wherever required, ensuring they are drawn with appropriate tools for precision.
- For numerical problems, **show each step clearly** with proper calculations, units, and explanations to ensure full transparency.
- Write your name, class roll no. section clearly on first page.

Question
1. Define Soft Computing. Outline its characteristics and application areas.
2. Compare areas where hard computing fails but soft computing succeeds, and justify.
3. With the help of a neat labeled diagram, explain the structure of a biological neuron. Describe its main elements and explain step by step how a neuron works to transmit information.
4. What is an Artificial Neural Network (ANN)? Describe its basic components including inputs, weights, bias, transfer function, and learning rate.
5. What are Activation Functions in neural networks? Write down the mathematical formulas for Identity Function, Step Function(binary & bipolar), Sigmoid Function(binary and bipolar), ReLU, and Tanh.
6. For the following sets of inputs and weights, calculate the net input (y_{in}) of the neuron and determine the output using the specified activation functions: 1. Case I (Binary Step, Binary Sigmoid and Bipolar Sigmoid activation function): • Inputs: $x_1=2.5$, $x_2=1.2$, $x_3=-0.5$, $x_4=3.0$ • Weights: $w_1=0.8$, $w_2=-1.5$, $w_3=0.3$, $w_4=1.1$ • Bias $b= -0.2$ 2. Case II (Tanh Activation Function): • Inputs: $x_1=-1.0$, $x_2=0.5$, $x_3=2.3$ • Weights: $w_1=2.0$, $w_2=-0.8$, $w_3=0.6$
7. Describe the different architectures of Artificial Neural Networks -Single-Layer Feedforward, Multilayer Feedforward, Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) and illustrate each with suitable diagrams.
8. Explain the McCulloch-Pitts (M-P) model of a neuron. Draw and discuss its architecture, working principle and limitations.
9. Differentiate between Supervised Learning, Unsupervised Learning, and Reinforcement Learning, explaining their key characteristics and applications.
10. Explain perceptron network with suitable diagram. Also, state the limitations of a single-layer Perceptron in solving real-world problems.

11. What is Backpropagation in neural networks? Explain its purpose and step-by-step algorithm for the Backpropagation learning method.

12. Implement the following logic using MP Neuron model:

Inputs Output

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Truth table 1

Inputs Output

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Truth table 2

Given the datasets below, how would you decide which learning technique to apply in each of the dataset and also clearly mention the dependent and independent variables?

DATASET A: Customer Churn (Telecom Industry)

Customer ID	Monthly Usage (GB)	Contract Type	Churn (Yes/No)
C001	12.4	Prepaid	No
C002	5.1	Postpaid	Yes
C003	8.9	Prepaid	No
C004	2.3	Postpaid	Yes

DATASET B: Student Performance (Education Sector)

	Student ID	Study Hours/Week	Attendance (%)	Passed (Yes/No)
S001	12	90	Yes	
S002	4	60	No	
S003	15	95	Yes	
S004	6	70	No	
S005	10	85	Yes	
S006	3	55	No	

DATASET H: Salary Prediction (HR/Job Analytics)

	Employee ID	Experience (Years)	Age	Education Level (1–3)	Salary (USD)
E001	2	24	1	32000	
E002	5	29	2	48000	
E003	8	35	2	62000	
E004	12	40	3	80000	
E005	15	45	3	95000	
E006	20	50	3	120000	

DATASET B: E-commerce Customer Behavior

Customer ID	Average Purchase Value	Time on Site (minutes)	Number of Visits
E101	120.5	45	5
E102	75.0	30	3
E103	210.0	60	6
E104	95.5	35	4

Explain the concept of perception and convergence rule. How this is implemented in perceptron network? Write its effectiveness in single layer perceptron network.

Discuss the effect of the learning rule coefficient (learning rate) on the performance of backpropagation networks. What happens if the learning rate is too high or too low?